

Jyothir Raghavalu Bhogi

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SUMMARY

AI Backend Engineer focused on building production-oriented AI services, LLM agent pipelines, and RAG systems using Python, FastAPI, LangChain, and REST APIs. Experienced in structured AI workflows, output validation, retrieval systems, API integration, testing, and Docker-based deployment, with a focus on reliable and maintainable backend systems.

EDUCATION

Lovely Professional University

B.Tech in Computer Science and Engineering

Jalandhar, Punjab

Aug 2022 – May 2026

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, NLP, Deep Learning, Data Structures, Algorithms, DBMS, Distributed Systems

EXPERIENCE

AI Intern

Know AI

June 2026 – Present

Ahmedabad, Gujarat

- Develop production AI services using Python, FastAPI, LLMs, RAG pipelines, and modern AI frameworks.
- Build REST APIs using FastAPI to expose AI capabilities through reliable backend services and workflows.
- Develop LangChain agent pipelines using structured outputs, embeddings, retrieval, and tool integration.
- Implement validation, testing, evaluation, experimentation, and error handling for reliable AI applications.
- Deploy and monitor backend services using Docker, Git, REST APIs, and production-oriented engineering workflows.

Data Engineer Intern

flickDone

June 2025 – Aug 2025

Remote (South Korea-based company)

- Built scalable Python and SQL data pipelines processing 50,000+ records for analytics and backend workflows.
- Developed ETL pipelines using Python, Pandas, and NumPy for data ingestion, validation, transformation, and processing.
- Built reliable data workflows supporting machine learning applications and structured analytical datasets.
- Implemented automated validation and processing workflows to improve data quality, consistency, and reliability.
- Contributed to REST APIs and CI/CD workflows using GitHub, Docker, and Linux development environments.

PROJECTS

Regulatory Compliance RAG Assistant | *Python, ChromaDB, BM25, Hugging Face*

Apr 2026 – Jun 2026

- Built RAG system using Python, ChromaDB, BM25, Hugging Face, and Streamlit for regulatory document processing.
- Implemented hybrid retrieval combining vector search and BM25 with Reciprocal Rank Fusion for reliable retrieval.
- Integrated cross-encoder reranking and LLM workflows to generate structured, context-aware responses with citations.
- Evaluated retrieval using Hit Rate and MRR across BM25, vector, hybrid, and reranked approaches.

AI Product Recommendation System | *Python, FastAPI, FAISS, LLMs*

Jun 2025 – Aug 2025

- Built an AI recommendation backend using Python, FastAPI, FAISS, embeddings, and LLMs.
- Implemented semantic search over SHL product metadata using BGE embeddings and FAISS retrieval.
- Integrated Claude 3 Haiku via OpenRouter to rerank retrieved products and improve response relevance.
- Developed REST APIs for natural-language queries, retrieval, reranking, and top-3 recommendations.

Morning Market Brief | *Python, FastAPI, LangChain, Docker*

Jun 2024 – Aug 2024

- Built a finance assistant using Python, FastAPI, LangChain, Streamlit, and Hugging Face for market updates.
- Designed multi-agent workflows for data retrieval, scraping, financial analysis, and response generation.
- Implemented FAISS and LangChain retrieval for financial insights with voice input and output capabilities.
- Containerized the application using Docker for reproducible development and backend deployment workflows.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++

GenAI: LLMs, Generative AI, RAG, Agentic AI, Agent Orchestration, Structured Outputs, Prompt Engineering

AI/ML: Machine Learning, NLP, Model Evaluation, Retrieval Systems, AI Workflows, Output Validation

Frameworks: FastAPI, LangChain, LangGraph, PyTorch, Scikit-learn, Hugging Face

Data & APIs: ChromaDB, Vector Databases, PostgreSQL, MySQL, Pandas, NumPy, REST APIs

Cloud & DevOps: AWS, Docker, Git, GitHub Actions, Linux, CI/CD, Deployment, Monitoring

Engineering: Backend Development, API Integration, Testing, Schema Validation, Workflow Automation, Production Systems